

P-256 global-row continuation: 64,000 new curves

Status: **complete**; verified negative frozen screen; solver advantage unset

Date: 2026-10-07

Outcome

The frozen continuation generated and independently replayed exactly **64,000 new P-256-isogenous curves** at global coordinates $0 \leq x < 1000$, $1000 \leq y < 1064$. Every emitted curve has a replayed incoming degree-11 or degree-13 kernel certificate. The exact streaming union of this strip with the immutable million-curve certificate contains **1,064,000 inputs and 1,064,000 distinct canonical 256-bit j-invariants**.

No frozen structural screen fired:

- `qr_prefix_64` ranged from 13 through 48, with zero values at the comparison threshold 52 or primary threshold 54;
- minimum canonical signed b length was 240 bits, above 224;
- minimum signed a length among models without $a = -3$ was 239 bits, above 224; and
- no j-invariant, canonical model, full ICV1 identity, 48-bit EC1 alias or full UID repeated inside the strip.

One expected 32-bit display-slug collision occurred inside the strip. It is not a full-identity collision. The cross-certificate auditor found no repeated full-width j-invariant.

The registered rule therefore selected no candidate for a factor-base test or whole index-calculus solve. **ECDLP speedup remains unset**.

Requirement-to-evidence accounting

requirement	status	acceptance evidence
continue beyond the million prefix	verified	64 complete new rows, global $y = 1000..1063$
preserve exact isogeny evidence	verified	64,000 incoming kernels, Velu codomains and target isomorphisms independently replayed
prove cumulative identity count	verified for j	union receipt reports 1,064,000 distinct canonical j values
retain frozen anomaly rules	verified	no adaptive threshold or detector change
establish an index-calculus speedup	not established	relation generation, matrix work, transport, recovery and matched rho were not run
exhaust 2^{32} or the full class	not established	cumulative coverage is 0.024773180% of 2^{32}

The evidence sources are the [frozen protocol](#), exact [generation receipt](#), independent [replay receipt](#), exact [j-union receipt](#), and typed [transfer assessment](#). The implementation is in [window.rs](#) and the [native CLI](#).

Construction and map boundary

The context curve at (0,999) was reconstructed from P-256 and was not counted again. The first emitted curve (0,1000) carries the new degree-13 certificate from that context. Every subsequent spine or row curve carries its ordinary incoming certificate. Generation used one deterministic order audit point per curve; replay used two fresh points beginning at $x = 7$.

The endpoint (999,1063) is connected to P-256 by a chain whose composite degree is $13^{1063} * 11^{999}$, a 7,390-bit integer. This is a compact chain representation. The experiment did **not** materialize a single large-degree kernel polynomial, a single rational map of that degree, or time payload transport through the complete chain.

Verified global-row continuation (representative endpoints)

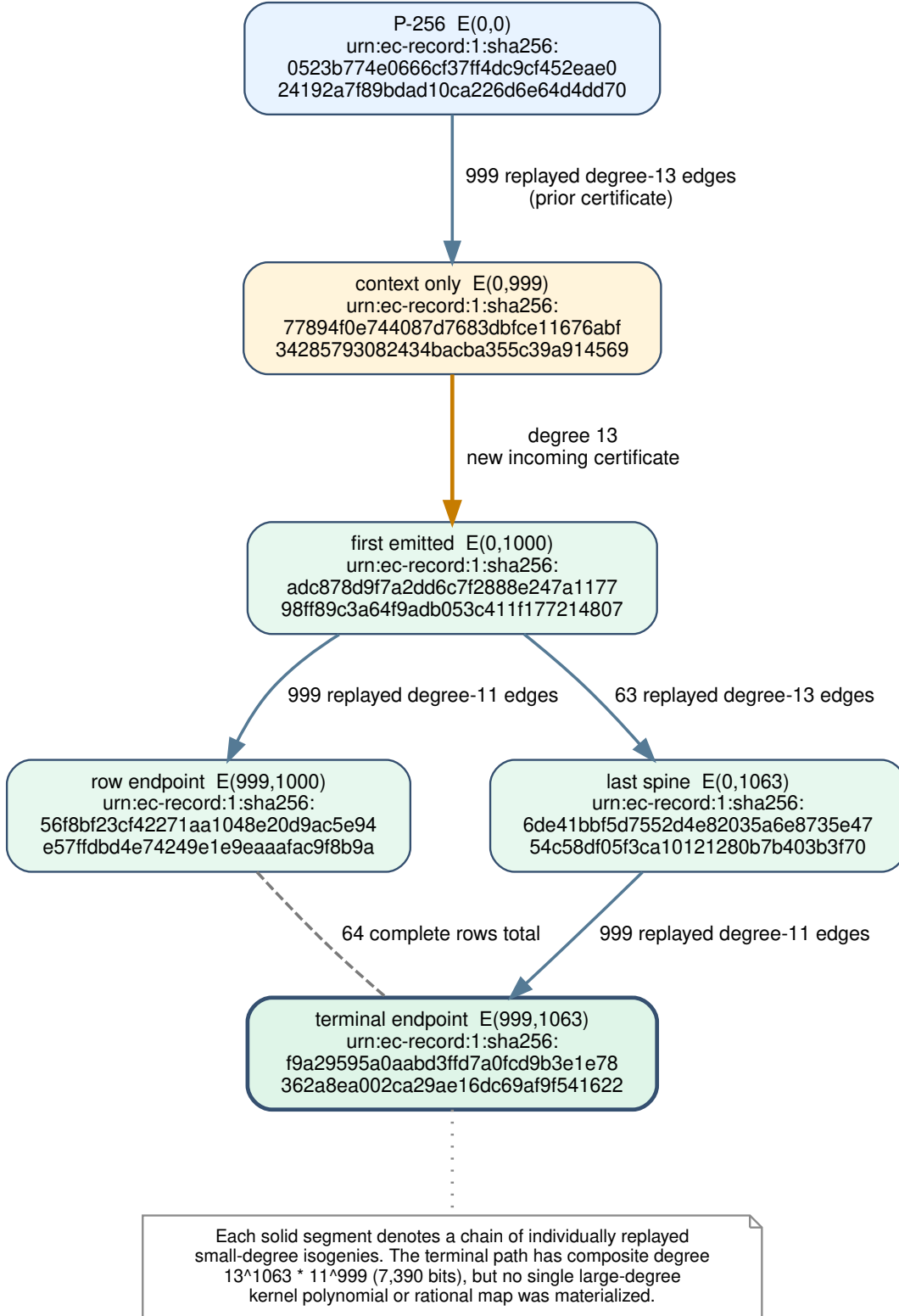


Figure 1: Representative verified lattice and full endpoint curve identifiers

Every small edge is a separable cyclic isogeny over F_p . Because 11 and 13 are coprime to the prime P-256 group order, the induced map is an isomorphism on the rational prime-order subgroup. That establishes subgroup preservation, not a lower discrete-log cost on the destination model.

Population and correctness gates

gate	generation	independent replay	result
new unique curves	64,000	64,000	pass
incoming parent edges	64,000	64,000	pass
kernel certificates	64,000	64,000	pass
prime-order audits	64,000 at one point	64,000 at two new points	pass
nonsingular / valid generator	64,000 / 64,000	same	pass
unique j / model / full ICV1 / EC1 / UID within strip	64,000 each	same	pass
exact uncompressed bytes	114,464,825	114,464,825	pass
strip record-chain SHA-256	full digest below	same	pass
union with million prefix	—	1,064,000 / 1,064,000 unique j	pass

The strip record-chain SHA-256 is 7d542b93aec2f21d9237e86f9d079cee4f858b5650bc0995c2f85ca0857cf87d.

The union reader verified the schema, terminal summary, curve count and rolling chain of each input before inserting canonical 256-bit j values. A focused fixture supplied the same certificate twice and was rejected with both duplicate coordinates, so overlap is a tested failure path rather than an assumed invariant.

Structural screens

statistic	frozen threshold	observed in new strip	decision
qr_prefix_64 primary	at least 54	maximum 48; zero hits	no candidate
qr_prefix_64 comparison	at least 52	zero hits	no candidate
canonical b signed bits	at most 224	minimum 240 at (209,1007)	no candidate
non-a=-3 signed a bits	at most 224	minimum 239 at (531,1047)	no candidate

The new strip's QR-count mean was 31.987078 and population variance was 16.380380. Combining its histogram with the prior million gives mean 31.996158 and variance 16.302705. These are descriptive calculations over the deterministic screen, not independent-sample inference.

The single value at 13 is a post-execution low-tail observation and was not a frozen trigger. Under only the simple Binomial(64, 1/2) reference, $P[X \leq 13] = 9.4048e-7$, or 0.0602 expected values in 64,000 trials; the probability of at least one is about 5.84%. The probes and isogenous models are not assumed independent, so this calculation neither establishes nor rules out a structural effect.

The strongest cumulative screen values remain in the prior million prefix: maximum QR count 51, minimum signed b length 233, and minimum non-a=-3 signed a length 236. The continuation does not change the frozen candidate decision.

Coverage boundary

population	exact unique curves	fraction of 2^{32}	coverage percent	ECDLP speedup
prior million prefix	1,000,000	0.0002328306437	0.023283064%	unset
new strip increment	64,000	0.00001490116119	0.001490116%	unset
exact cumulative j union	1,064,000	0.0002477318048	0.024773180%	unset

population	exact unique curves	fraction of 2^{32}	coverage percent	ECDLP speedup
requested boundary	4,294,967,296	1	100%	unset

The cumulative count is one part in 4,036.623398 of 2^{32} . The frozen protocol printed 4,036.624060 in its planned reciprocal row; that was an arithmetic transcription error. Its exact count and percentage were correct, and the frozen file has not been rewritten after seeing results.

Neither grid coordinates nor path degree define all distinct curves in the P-256 isogeny class. This is exact coverage of two replayed certificates, not an exhaustive class enumeration and not evidence about unexecuted coordinates.

Execution receipt

The clean implementation commit bound into the new certificate is 581aeff3d89b412e718207e05f4002808271f1ef. The release binary was 4,874,560 bytes with SHA-256 1591684928920cc67e7be5b44d50d76b8489560189ef59d072faa8800be1d8c3.

phase	threads	wall	user CPU	system CPU	peak RSS
generation, one audit point	4	369.899 s	1,380.471 s	0.300 s	86,233,088 B
replay, two audit points	5	411.530 s	1,409.073 s	0.145 s	88,707,072 B
two-certificate j union	1	19.514 s	19.588 s	0.416 s	183,721,984 B

Host: Linux 6.18.44 x86-64 KVM, five exposed AMD EPYC 9V74 cores, 18,882,699,264 bytes RAM; Rust 1.98.0 (88d9e12ae 2026-08-18).

```
p256_isogeny_million --threads 4 generate-strip \
  --width 1000 --y-start 1000 --height 64 --batch-rows 8 \
  --audit-points 1 \
  --source-commit 581aeff3d89b412e718207e05f4002808271f1ef \
  --output p256-strip-y1000-h64.jsonl.gz
```

```
p256_isogeny_million --threads 5 verify-strip \
  --input p256-strip-y1000-h64.jsonl.gz \
  --audit-points 2 --audit-seed-x 7 --batch-rows 8
```

```
p256_isogeny_million audit-j-union \
  p256-grid-1m.jsonl.gz p256-strip-y1000-h64.jsonl.gz
```

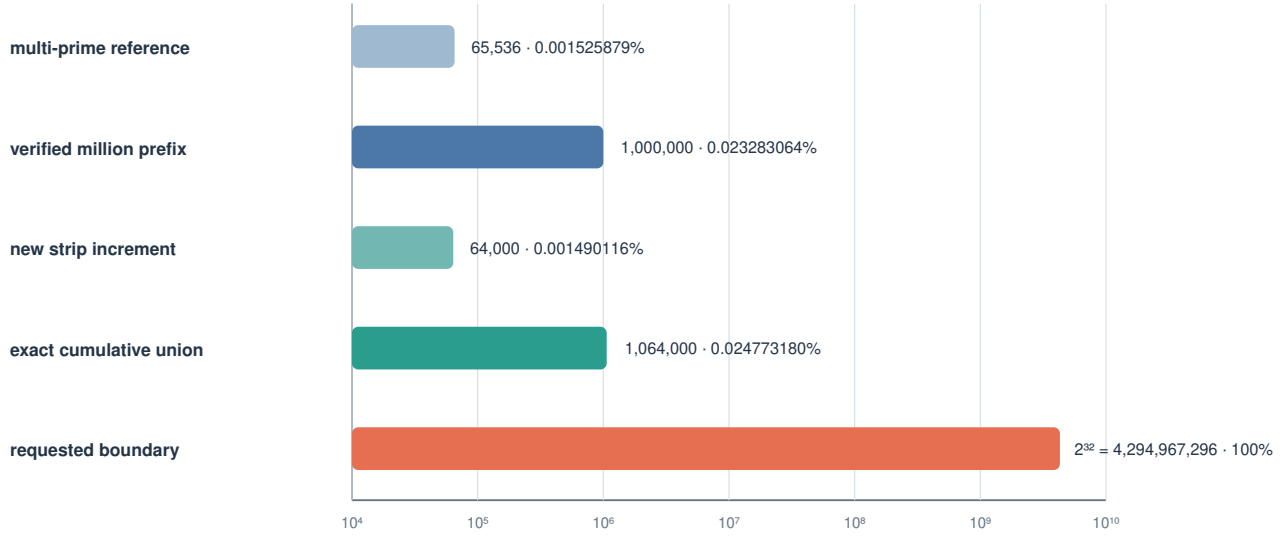
artifact	bytes	SHA-256
new canonical gzip	33,739,337	90f22f7aadaef78cff5a72889e5539b2ae81459d57d7a84b37a9009e96a3f701
new exact decompression	114,464,825	a24593f9d1d1c880b63f1da86d10e375981f72aaa358e1eae428e63b3d4714b1
generation receipt	4,921	f8965169b464dc8db294be0534d04c1963d2022ca2a7ef2049ffcd5d4f868305
replay receipt	4,963	a11ab708d6b3192faf1a4052ef2820bd8c6a43ffe14540af52b54970f8f26619
union receipt	1,654	44f5e746668bb797d51bcbcc26a795c1b32e83dae8c96dcd877167b97de086bb

Full SHA-256 digests:

- canonical gzip: 90f22f7aadaef78cff5a72889e5539b2ae81459d57d7a84b37a9009e96a3f701
- exact decompression: a24593f9d1d1c880b63f1da86d10e375981f72aaa358e1eae428e63b3d4714b1
- generation receipt: f8965169b464dc8db294be0534d04c1963d2022ca2a7ef2049ffcd5d4f868305
- replay receipt: a11ab708d6b3192faf1a4052ef2820bd8c6a43ffe14540af52b54970f8f26619
- union receipt: 44f5e746668bb797d51bcbcc26a795c1b32e83dae8c96dcd877167b97de086bb

Verified P-256 screening coverage against 2^{32}

$\log_{10}$ x-axis; counts are exact within replayed certificates, with no sampling error bar.



The cumulative audit covers $1 / 4,036.623398$ of the requested boundary; it is not an exhaustive class search.

Figure 2: Log-scale exact coverage compared with the requested 2^{32} boundary

The large certificate remains local at `/tmp/p256-j-strip-20261007/runs/p256-strip-y1000-h64/`. A local path is not durable publication. No S3 upload or Cairn network submission was attempted because this environment has no configured ambient AWS identity, destination bucket, Cairn objective or signing identity. The implementation can stage and validate content-addressed storage once those are provided through the worker environment.

Validation and retained failures

- focused generation/replay, deterministic identity, boundary-tamper and parent-tamper tests passed;
- adjacent small strips replayed and unioned to 20 distinct j values;
- a deliberately overlapping small union failed with both locations;
- the legacy small-grid replay/tamper test passed;
- native check and clippy with warnings denied passed;
- focused Rust formatting and git diff –check passed before production.

Two pre-production failures are retained rather than overwritten:

1. a debug build exhausted its disposable target filesystem and failed with No space left on device; the disposable target alone was removed, then rebuilt with incremental compilation and debuginfo disabled; and
2. the first production launcher used unavailable `/usr/bin/time` and exited 127 before the program started. Its empty receipt and stderr remain beside the successful attempt locally.

Neither failed attempt is counted as a passing run.

Scoreboards and conclusion

Canonical ECDLP and solver scoreboards remain unchanged because this enumeration measured no relation yield, linear algebra, target-log recovery, matched Pollard rho or end-to-end time. The only quantitative additions are curve coverage, construction/replay cost and exact union identity count.

Within the executed continuation, there is no frozen structural candidate and no full-width j discrepancy. The result does not support the claim that index calculus is faster on an isogenous P-256 model.

The next defensible step is another disjoint, preregistered global-row strip, followed by the same replay and union gates.

A solver experiment should begin only if a frozen detector selects a curve; then it must compare cold, full-pipeline index calculus with matched Pollard rho while including map transport and recovery costs.